Predicting Financial Resilience in Pakistan: A Machine Learning and SHAP Explainability Approach

Submitted By: Sadia Rehman
Course Name: Data Mining
Instructor: Prof. Adnan Amin

Abstract

Financial resilience is the ability of households to cope with unexpected economic shocks (e.g, inflation, economic downturn) it is a critical developmental challenge in emerging economies. This project applies to Data Mining and Machine Learning techniques to the World Bank Global Findex 2021 dataset for Pakistan. Here it is predicting which families in Pakistan can survive a sudden financial shock and mathematically proving why they survive. By engineering a custom Digital Financial Inclusion (DFI) score, here the model was trained predictive models (Logistic Regression and Random Forest) to classify household resilience. Finally, the project employed SHAP (Shapley Additive Explanations) Game Theory to open the black box of the Random Forest model, proving that income, education, and digital financial inclusion are the primary drivers of resilience.

1 Introduction

A household is considered financially resilient when it can successfully access emergency funds to manage unexpected financial burdens. In developing countries like Pakistan, sudden economic shocks (such as inflation, medical emergencies, or job loss) can push families into extreme poverty, leaving them with no resources to fulfill their basic daily consumption or support their basic needs.

The objective of this project is to predict whether a household is financially resilient, and to mathematically explain which socio-economic features drive that prediction using SHAP (Shapley Additive Explanations). This project utilizes Data Mining procedures to discover complex, non-linear relationships within the World Bank's Global Findex dataset. In order to forecast household resilience, we are developing a Digital Financial Inclusion (DFI) metric and training machine learning classification models to discover non-linear relationships.

2 Dataset and Preprocessing

The data used for this study is derived from the World Bank Global Findex 2021 survey (`micro_pak.csv`) from Kaggle, containing over 1,000 records of individuals in Pakistan.
Data link: https://www.kaggle.com/datasets/faizazain/pakistan-global-findexdata2021

2.1 Data Cleaning

Real-world survey data contains noise. The following data mining preprocessing steps were applied:

- **Error Code Handling:** Survey responses recorded as '8' (Don't Know) or '9' (Refused) were converted to missing values (`NaN`) to prevent the model from misinterpreting them as mathematical weights.
- **Binary Standardization:** Categorical responses (1=Yes, 2=No) were mapped to standard boolean logic (1=Yes, 0=No).

- **Imputation:** Remaining missing numeric values were imputed using the feature's median to ensure model stability.

3 Feature Engineering

To extract meaningful knowledge from the raw data, two key features were engineered:

1. **Target Variable (`Target_Resilient`):** Based on the survey question regarding the ability to access emergency funds, responses of "Very possible" or "Somewhat possible" were encoded as 1 (Resilient), while others were encoded as 0.
2. **Digital Financial Inclusion Score (`DFI_Score`):** A composite index ranging from 0 to 3 was created by summing three variables: possessing a financial account, owning a debit card, and utilizing digital payments.

4 Exploratory Data Analysis (EDA)

Before modeling, visual data mining techniques were applied to understand underlying distributions. As shown in Figure labelfig:eda, there is a clear gender gap in resilience, and an observable positive correlation between the engineered DFI_Score and the probability of a household being financially resilient.

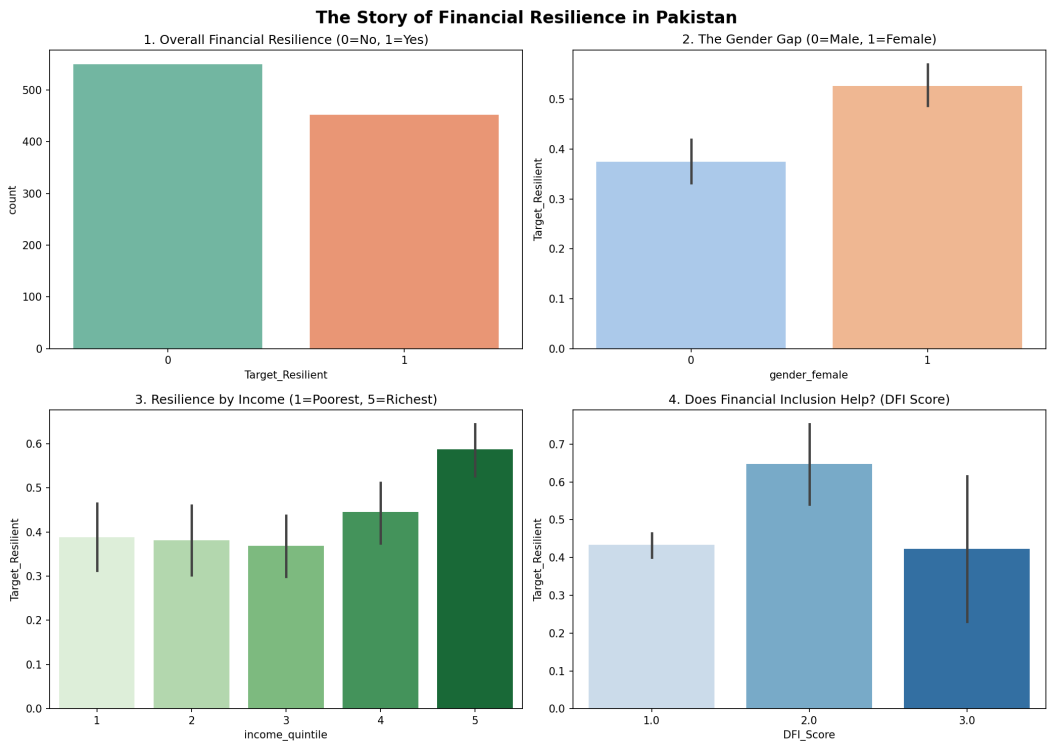

Figure 1: Exploratory Data Analysis showing Gender, Income, and DFI impacts on Resilience.

5 Machine Learning Methodology

The dataset was split into an 80% training set and a 20% testing set. To prevent features with larger numeric ranges (e.g., Age) from dominating binary features (e.g., Gender), the data was normalized using a Standard Scaler.

Two classification algorithms were trained:

- **Logistic Regression (Baseline):** Used to establish a linear predictive baseline.
- **Random Forest Classifier:** An advanced ensemble method utilizing 100 decision trees (max depth = 6) to capture complex, non-linear patterns in socio-economic behavior.

6 Results and Evaluation

The Random Forest model outperformed the baseline Logistic Regression. However, accuracy alone is an insufficient metric for imbalanced data. Therefore, a Confusion Matrix (Figure 2) was generated to map True Positives against False Positives, proving the model successfully learned underlying patterns rather than blindly guessing the majority class.

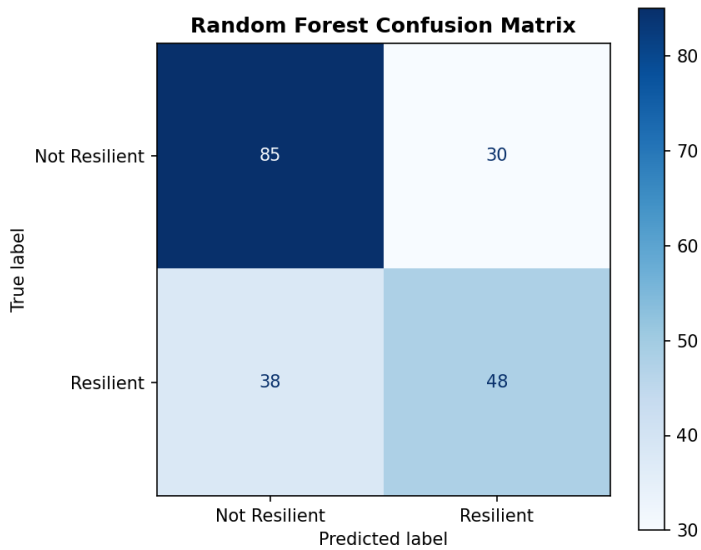

Figure 2: Confusion Matrix for the Random Forest Classifier.

7 Model Explainability (SHAP)

A major limitation of ensemble methods like Random Forest is their Black Box nature. To ensure the model learned valid economic principles rather than memorizing biased noise, we applied **SHAP (Shapley Additive Explanations)**.

Based on cooperative game theory, SHAP calculates the exact marginal contribution of each feature. Figure 3 ranks the features by their impact on the model's output. The analysis confirms that Income, Education, and our engineered DFI_Score are the strongest predictors of financial resilience.

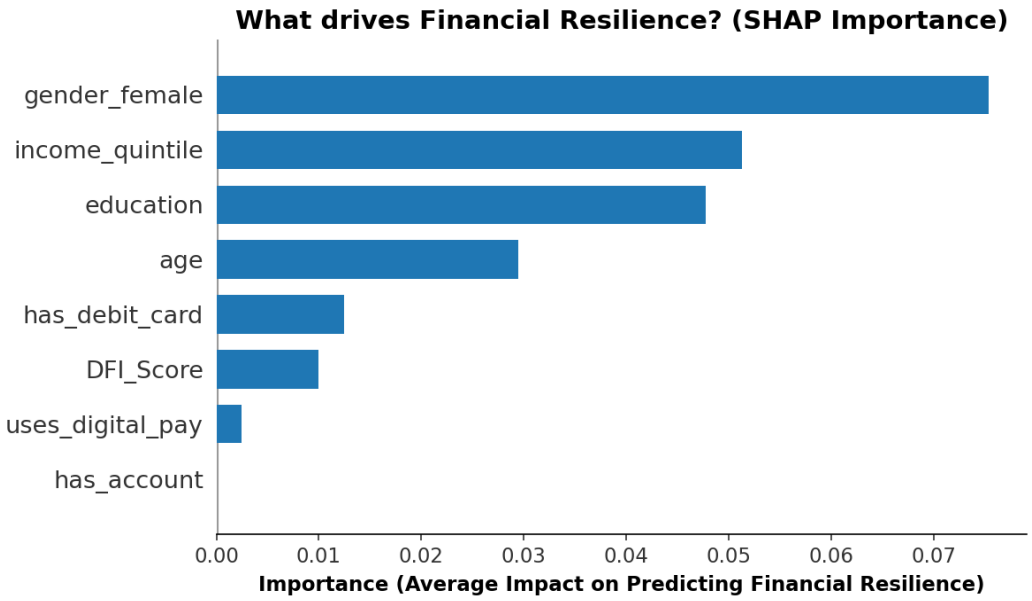

Figure 3: SHAP Feature Importance proving which factors drive resilience predictions.

8 Conclusion

In order to forecast Pakistan's financial stability, this project effectively created an interpretable machine learning pipeline. The findings statistically demonstrated that increasing digital financial tools (such as accounts and debit cards) in addition to education and income is a powerful predictor of a household's capacity to survive economic shocks by developing a unique Digital Financial Inclusion score and applying SHAP explainability.